SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **CSA63**

Course Name: **Threat Intelligence and Network Security**

Duration: **1: 30 Hr**

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Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.
CO2	Identify common network attacks and analyze threat intelligence sources and data collection techniques.

Answer all the Questions

1. Medilife Hospital runs a patient-records portal that doctors, nurses and insurers access over the hospital network. In one week the security team logs three incidents: (i) ransomware encrypts the radiology database and staff cannot open scans, (ii) a disgruntled employee silently changes a patient's blood-group entry in the lab system, and (iii) an attacker captures unencrypted patient data travelling on the Wi-Fi network without altering it. Interpret each incident against the CIA triad, identifying the security goal that is primarily violated and justifying your choice. Classify each incident as a passive or active threat and as an internal or external threat. Recommend one preventive control for each incident, explaining how it restores the violated goal. Then analyse how the three goals can conflict with one another when the hospital designs access to emergency patient records, and explain why a balance among confidentiality, integrity and availability is required for a hospital network.

(10) (CO1) (BL3)

2. The Security Operations Centre (SOC) of a national bank receives four intelligence items in a single morning: (i) a quarterly report from a consulting firm stating that financially motivated groups are expected to target mobile-banking customers over the next year, (ii) a list of malicious IP addresses and domain names linked to a phishing campaign, (iii) an advisory describing the attacker's planned timing, tools and likely targets for an imminent attack on the bank's payment gateway, and (iv) a write-up of the attacker's tactics, techniques and procedures (TTPs) in a recent banking-malware campaign. Classify each item as strategic, tactical or operational threat intelligence, justifying your choice. Compare the three types of threat intelligence in terms of

   (a) target audience and decision level,

   (b) time horizon and level of technical detail, and

   (c) typical sources and the action taken by the bank.

   Finally, analyse what the bank would lose if it relied only on tactical intelligence (IoC lists) and ignored strategic and operational intelligence.

   (10) (CO1) (BL4)

3. a) An e-commerce company wants to build a threat intelligence programme after a data breach. Apply the threat intelligence lifecycle to the company by describing, for each phase (direction/planning, collection, processing, analysis, dissemination and feedback), the activities carried out and the output produced. Explain how the feedback phase keeps the lifecycle continuous, and give two applications of threat intelligence in the IT industry (such as SOC operations, incident response or vulnerability prioritisation) that the company can adopt.

   (5) (CO1) (BL4)

   b) An employee in the finance department receives an email that appears to come from the CEO, urging an immediate payment and containing a link to a "secure invoice portal". The employee clicks the link, enters credentials and downloads an attachment, after which files on the shared drive become inaccessible. Analyse the incident by identifying the types of attack involved (phishing and malware), the stages of the attack from email delivery to impact, and the indicators in the email that should have raised suspicion. Suggest two technical and two user-awareness measures to prevent a recurrence.

   (5) (CO2) (BL4)

4. A university campus network experiences three problems: (i) the online examination portal becomes unreachable when thousands of requests per second arrive from many compromised devices around the world, (ii) students connected to the free campus Wi-Fi are redirected to a fake login page, and their traffic is relayed through an attacker's laptop that reads and modifies the data, and (iii) an attacker sends packets with a forged source address of the registrar's server to gain access to the administrative VLAN, while another user runs a packet-capture tool to read unencrypted traffic. Identify the attack in each case and explain how it works. Distinguish between DoS and DDoS attacks, between spoofing and sniffing, and between sniffing (passive) and a man-in-the-middle attack (active). For each attack, describe one detection technique and two countermeasures (for

example, rate limiting and traffic scrubbing, HTTPS/TLS with certificate validation, ARP inspection, encryption and ingress filtering) that the university should deploy. Also explain how a vulnerability assessment of the campus network would help reduce the likelihood of these attacks.

(10) (CO2) (BL4)

5. A threat-intelligence analyst at a managed security provider receives a commercial threat feed containing the following entries related to a ransomware campaign: IP address 203.0.113.45, domain "secure-login-update[.]com", file hash (SHA-256) of a malicious dropper, URL "hxxp://secure-login-update[.]com/invoice.exe", and the email subject line "Urgent: Invoice Overdue". The provider also wants to supplement the feed with open-source intelligence (OSINT). Analyse the feed by classifying each entry as a network, host or email indicator of compromise (IoC) and evaluating which entries are easy or hard for an attacker to change. Explain how the analyst should assess the quality of the feed (timeliness, accuracy, relevance and context) before deploying the IoCs to firewalls and SIEM tools, and the risk of false positives. Identify four OSINT sources and the data collection techniques (such as monitoring social media, paste sites, public breach databases, domain WHOIS and passive DNS) that can enrich the feed, and compare OSINT with commercial threat feeds in terms of cost, reliability and coverage. Finally, justify which combination of sources the provider should use.

(10) (CO2) (BL4)

CSA6303 Threat Intelligence and Network security.

CIA 1

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Q1: Medilife hospital case study.

47

i) Ransomware encrypts radiology database.

Availability: The database is rendered inaccessible to authorized staff halting diagnostic workflows.

Active threat: Modifies state encrypts files and disrupts system operations

External threat: Typically originates from external malware authors / threat.

Backups: Restores availability by providing clean unencrypted data points for rapid recovery without paying ransome.

ii) Disgruntled employee alters blood-group entry.

INTEGRITY: Patient data was unauthorized modified, corrupting the accuracy and trustworthiness of medical record

ACTIVE THREATS: Deliberally modifies database records.

INTERNAL THREATS: Executed by an authorized insider leveraging internal access.

ACCESS CONTROL: Restores integrity by requiring supervised verification for critical data changes and preventing

unathorized edits.

iii) Attackers confidentiality Passive threat:

- Capturing unencrypted data over Wi-Fi

Patients records were exposed and read by unathorised third party.

- Intercepts / eavesdrops on traffic without altering or disrupting the network.

UNAUTHORIZED THREAT: Originates from an evandropper listning on the wireless medium.

ENCRYPTION: Restores confidentiality by encrypting data - in transit, making captured packets unradable ciphertext.

CONFIDENTIALITY Vs AVAILABILITY: Strictu confidentiality controls factor authentication slow down access speed. Delays in accessing allegy or blood-type data can lead to fatal.

INTIGRITY Vs AVAILABILITY: Intigrity is ensured by cryptographic techniques but jopardizes immediate availability during critical trauma interventions.

Necessity of balance in a healthcare network.

- Confidentiality - Trust collapses and legal liabilities
- Intigrity - Corrupted data resulting in fatal outcome.
- Availability - access fail during critical critical operations.

Question 2:

1. Quarterly report on mobile banking market: High level trend analysis focused on board business risk, long-term threat actor motivation and financial sectors over a multi-month

2. List of malicious IP address and domain names: Specific, short-lived indicators of compromise (IoCs) used directly by technical systems for immediate threat feed.

3. Advisory on attack timing, tools: Actionable details an incoming/imminent specific campaign, including specific adversary capabilities, timing and direct targets.

4. Write-up of TTP's from recent malware: Detailed behavioral methodologies mapped to framework like MITRE Attack, focusing on how attackers operate rather than just static IoCs.

ASPECT	STRATEGIC INTELLIGENCE	OPERATIONAL INTELLIGENCE	TACTICAL INTELLIGENCE.
a) Target audience & decision level	C-suite, Board of directors, CISO's	SOC Mangers, Incident respons lead, Threat hunt	Security analyst, firewall / SIEM Administrators.
b) Time Horizon and Detail level	long term; low technical details high business, risk context	Medium-term; High contextual detail on adversary TTPs and attack plans	short-term; High technical details, very long contextual logevitity.

c) Industry reports.	Sh government advisor threat Budget and allocations, policy updates long-term security arch	Dark, Web monitoring. Proactive threat hunting, rule creation for specific TTPs	ISAC, automati sandbox analysis. Block IP's domains on firewalls, update SIEM correlation rules.

Risks Relying solely on Tactical intelligence (IoCs)

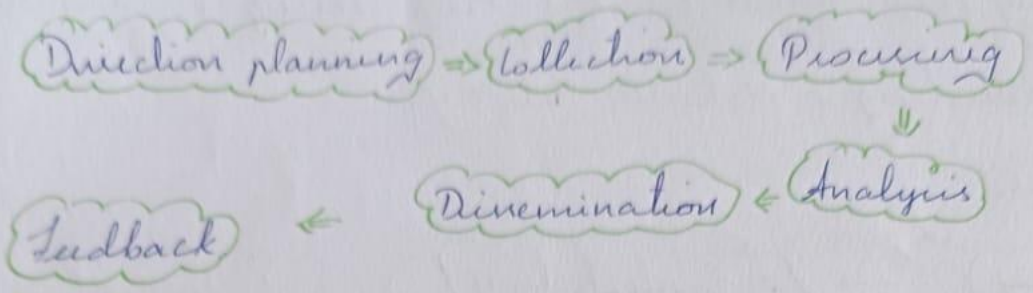
- Easily evade by adversaries: Attackers change IP address, domain names, and file hashes in seconds at zero cost

- Alert fatigue & High noise: Inundates SOC analysts with thousands of ephemeral indicators without providing context on priority or relevance.

- Loss of strategic foresight: The bank fails to anticipate emerging threats, structural security gaps, or strategic shifts in attack playbooks, leaving them purely reactive rather than proactive.

Question 3: Threat intelligence lifecycle

a) Threat intelligence lifecycle & Industry applications



i) Direction and planning: Define intelligence requirement based on business assets

ii) Collections: Gather raw data from internal SIEM logs, OSINT feeds, dark web forums, commercial vendors

iii) Procesing: Clean, normalized, parse and structure raw data extracting IP lists from text

iv) Analysis: Evaluate structured data for relevance, context and actionable insight to turn it into actionable

v) Dissemination: Distribute finished intelligence to the appropriate target audience.

vi) Feedback: Phase review the utility of disseminated intelligence with stakeholders to refine future

Industry application:

Vulnerability Prioritization

SOC operation & Automated blocking.

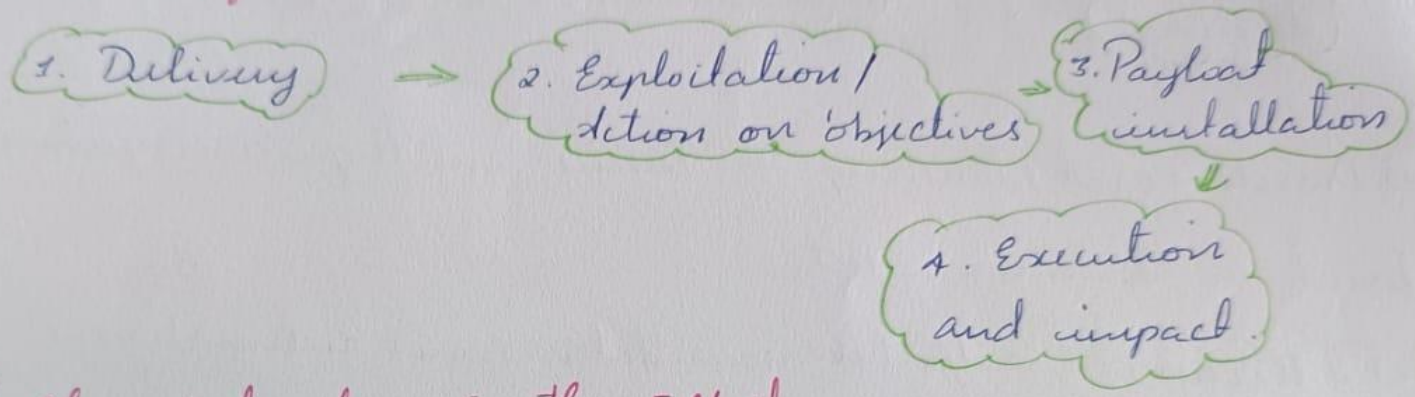
b) E-commerce Phishing Incident Analysis

Attack involved

Spear phishing / CEO fraud (social engineering)

Credential Harvesting & Malware delivery.

Attack stages.



Red flag indicators in the E-Mail.

- Urgency & Pressure Tactics
- Process Deviation
- Domain maintainance / his URL

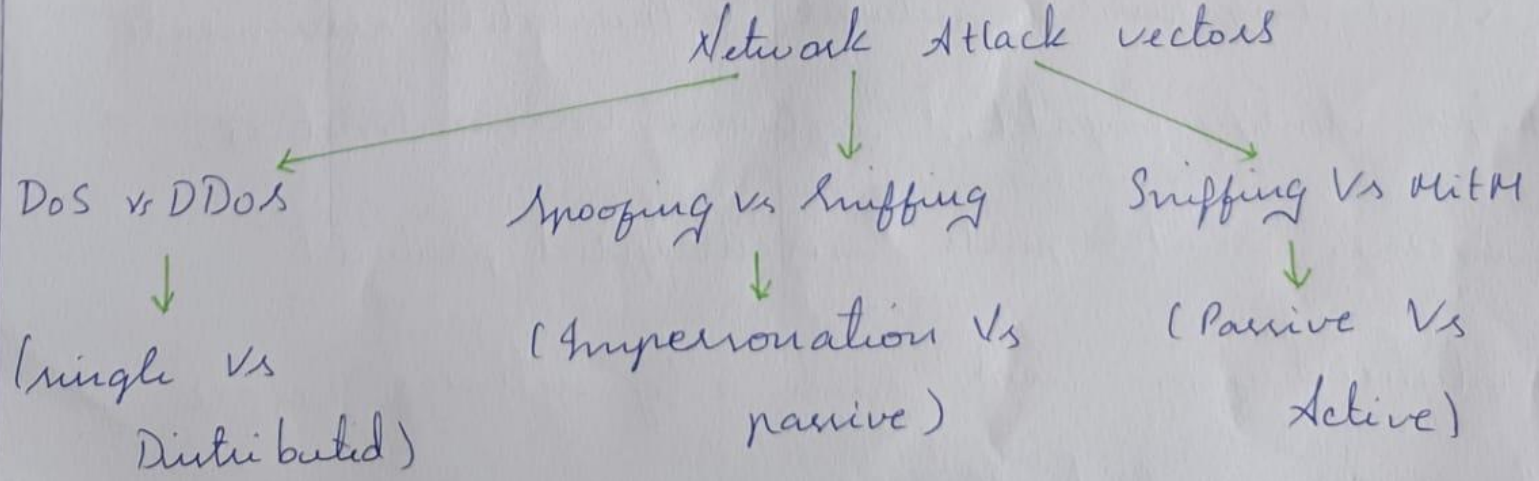
Prevention measures:

TECHNICAL MEASURES: Deploy email authentication protocols to block spoofed external sender domains.

USER-AWARENESS MEASURES: Conduct regular, randomized Phising simulation Training targeting high-risk department.

Question 4: University network Attack & Defense analysis.

	Incident	Indentified attack	& attack mechanism.
i	Exam portal unreachable via global request	Distributed Denail of service	A botnet of geographically dispersed compromised devices floods the server with traffic.
ii	Students redirected to fake login over public WiFi	Man in the middle	Attacker sets up a rogue AP or spoofs ARP response, interposing their device in the communication path to inject
ii	Packets with forged server IP / Traffic capture	IP spoofing & packet sniffing	Attacker crafts IP packets with fake source headers to bypass IP-based ACL's.



	Attack	Detection Techs	Countermeasure 1	Countermeasure 2
i	DDoS attack	Anomaly-based NetFlow / IPFIX monitoring for sudden traffic spikes	Restricts maximum incoming connections per client	Reroute network traffic through mitigation centers to filter out malicious packets.
ii	MitM / Rogue AP	Wireless Intrusion Detection system detecting unauthorized BSSIDs	Forces strict certificate validation & encrypted channels	Prevent ARP poisoning on switch ports
iii	IP spoofing & sniffing	Promiscuous mode detection tools on switch ports i allot on internal IP origins coming from physical interfaces	Drops packets whose source IP does not belong to the attached subnet.	Encrypt network layer payload and authentication switch access.

Conclusion:

⇒ Conducting periodic network vulnerability assessments helps identify unpatched software vulnerabilities, misconfig, switch ports, unencrypted protocol implementation and open ecly setting before adversaries exploit-them.

Question 5: Threat Indicator Analysis & OSINT Evaluation.

Indicator entry	IoC category	Attacker Situation Difficulty.
203.0.1135.45	Network IoC	Easy
secure-login-update.com	Network IoC	simple / medium
5th hash (SHA-256)	Host IoC	Trivial
hxxp://secure-login.../invoice.exe	Network / Host IoC	simple.

Threat feed quality Assessment.

Timeliness ⇒ Accuracy ⇒ Relevance ⇒ Context

Risk of false positive.

Opensource Intelligence Analysis

1. Public Code / Paste site: scraping public repositories for leaked API keys, credentials and published exploit scripts.

2. Domain Whois & Passive DNS Data: Querying historical domain registrants and IP resolution records to map out threat actor infrastructure network

3. Public Breach & File repositories: Virus Total, Abuse IPDB

4. Social Media & Security Blogs: Tracking real-time vulnerability disclosures, zero-day research releases and threat actor chatter

Hybrid approach:

Commercial Feeds: supply vetted, structured, high-accuracy tactical IoCs

OSINT provides broad, early-stage visibility into emerging zero days.